Exploiting Spatial and Color Symmetry in Equivariant Value Networks for the 2×2×2 Pocket Cube

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Abstract

Learning value functions for combinatorial search problems requires models that generalize reliably across symmetry-equivalent states. The 2×2×2 Pocket Cube possesses two physically meaningful symmetry groups: the 24-element rotation group $\mathcal{O}$ (spatial orientations) and the 720-element symmetric group S_6 (color relabelings), whose direct product $\mathcal{G} = \mathcal{O} \times S_6$ has 17,280 elements. We design and analyze two group-equivariant architectures: EMLP (equivariant to $\mathcal{O}$, ∼42K parameters) and EMLP-Col (equivariant to the full $\mathcal{G}$, ∼39K parameters), compared against a parameter-matched unconstrained MLP (∼40K parameters) as the primary baseline. We prove that both architectures are exactly invariant to their respective symmetry groups and show that equivariance to $\mathcal{G}$ implicitly provides 17,280-fold data augmentation. Empirically, at equal parameter count, EMLP-Col achieves a greedy solve rate of 97.2% at scramble depth 10 and a mean absolute error of 0.93 moves, compared to 94.2% and 1.24 moves for the spatial-only EMLP, and a solve rate of [pending] for the unconstrained MLP. The results show that building in symmetry directly—rather than leaving it for the model to learn from data—produces more consistent value landscapes that are critical for reliable beam-search guided solving.

1 Introduction

Combinatorial puzzles offer a clean testbed for evaluating how well neural networks exploit problem structure. Unlike most learning tasks where symmetry is approximate or data-driven, combinatorial puzzles often have exact, group-theoretic symmetry baked into their definition—symmetry that, if ignored, forces a model to rediscover the same relationships from data across every equivalent configuration.

The 2×2×2 Pocket Cube (Figure 1) is an especially attractive benchmark: its state space of 3,674,160 reachable configurations is small enough to solve exactly via breadth-first search (BFS), yielding ground-truth distances for every state [1]. A value network $f : \mathbb{R}^{144} \to \mathbb{R}$ that predicts BFS distance can guide greedy or beam search to solve scrambled cubes without tree search or explicit rollouts.

Two symmetries of the cube. The cube has two physically meaningful symmetry groups:

- **Spatial rotations** $\mathcal{O}$ (order 24): rotating the entire physical cube does not change its distance to the solved state. Any value function must satisfy $f(\pi \cdot \mathbf{x}) = f(\mathbf{x})$ for all $\pi \in \mathcal{O}$.
- **Color permutations** S_6 (order 720): systematically relabeling all six colors yields an isomorphic puzzle with identical structure. Thus $f(\sigma \cdot \mathbf{x}) = f(\mathbf{x})$ for all $\sigma \in S_6$.

[Figure: 2×2×2 cube with sticker labels]

Figure 1: The 2×2×2 Pocket Cube. Each of the 24 sticker positions is assigned one of 6 colors. The solved state has each face showing a single color.

Together these form the group $\mathcal{G} = \mathcal{O} \times S_6$ of order $24 \times 720 = 17{,}280$. An unconstrained MLP must learn all 17,280 equivalent views from data; an equivariant model encodes them exactly by construction.

Beam search as the evaluation criterion. A value network is only useful insofar as it guides a solver to the solved state. Recent work by Chervov et al. [2] (the CayleyPy project) demonstrates that beam search guided by a learned distance heuristic is the most effective strategy for solving Rubik's cubes of all sizes, achieving 98.4% optimality on the standard 3×3×3 cube and surpassing all prior methods—including DeepCubeA [1] (60.3% optimality) and EfficientCube [11] (69.6% optimality)—while being 26× faster. Crucially, Chervov et al. characterize the symbiotic role of the two components: *"The neural network provides heuristics on where to make the next steps, while the graph pathfinding technique compensates for any possible incorrectness in the neural network predictions."* This framing directly motivates our evaluation strategy: we measure value network quality through beam search solve rate, because a good heuristic should produce a reliable value landscape for local move selection, even under a modest beam width.

Contributions. We make the following contributions:

1. We design a **Color-Equivariant Convolution layer** (ColorConv) that is jointly equivariant to $\mathcal{G} = \mathcal{O} \times S_6$, using a block-of-six channel structure to represent S_6 acting on the color dimension.
2. We provide **formal proofs** of equivariance for both the spatial-only layer (SpatialConv) and the combined layer (ColorConv), and a corollary establishing invariance of the complete networks.
3. We show that equivariance to $\mathcal{G}$ is equivalent to **implicit** 17,280**-fold data augmentation**—a strictly stronger guarantee than explicit augmentation.
4. We demonstrate empirically that at equal parameter count (∼40K), the equivariant models substantially outperform an unconstrained MLP on every metric, and expose the **"local consistency gap"**: symmetry matters not just for average prediction accuracy but for the per-move ranking consistency that beam search depends on.

2 Background

2.1 The 2×2×2 Pocket Cube

The 2×2×2 cube has 6 faces, each divided into 4 stickers, giving 24 sticker positions. We index positions $i \in \{1, \ldots, 24\}$ and colors $c \in \{1, \ldots, 6\}$. A state $\mathbf{x} \in \{0, 1\}^{24 \times 6}$ is a one-hot matrix where

$x_{i,c} = 1$ if sticker i shows color c. We flatten $\mathbf{x}$ to a vector of dimension 144 as network input.

The move set consists of six quarter-turn moves $\{R, R', U, U', F, F'\}$ in the quarter-turn metric (QTM). BFS over the Cayley graph of all 3,674,160 reachable states establishes God's number: the maximum BFS distance is $d_{\max} = 14$ in QTM [9, 10]. The exact distance $d(\mathbf{x}) \in \{0, 1, \ldots, 14\}$ for every state can be precomputed and stored in a lookup table, providing noiseless supervision for training a value network $f(\mathbf{x}) \approx d(\mathbf{x})$.

2.2 Group Theory

Definition 2.1 (Group action)**.** *A group G* acts *on a set X if there is a map $G \times X \to X$, $(g, x) \mapsto g \cdot x$, satisfying $e \cdot x = x$ and $(gh) \cdot x = g \cdot (h \cdot x)$ for all $g, h \in G$ and $x \in X$.*

Definition 2.2 (Orbit)**.** *The* orbit *of $x \in X$ under G is $G \cdot x = \{g \cdot x : g \in G\}$. Two elements $x, x' \in X$ lie in the same orbit iff $\exists g \in G$ such that $g \cdot x = x'$.*

The rotation group $\mathcal{O}$. The rotation group of the cube, $\mathcal{O}$, has 24 elements. Each $\pi \in \mathcal{O}$ acts on sticker positions as a permutation $\pi : \{1, \ldots, 24\} \to \{1, \ldots, 24\}$. The induced action on states is

$$[\pi \cdot \mathbf{x}]_{i,c} = x_{\pi^{-1}(i),\, c}. \tag{1}$$

We represent $\mathcal{O}$ by its Cayley generators: a 90° rotation around the right-left axis (R_{90}) and around the up-down axis (U_{90}).

The color permutation group S_6. The symmetric group S_6 consists of all $6! = 720$ permutations of the six colors. Each $\sigma \in S_6$ acts on states by permuting the color channels:

$$[\sigma \cdot \mathbf{x}]_{i,c} = x_{i,\, \sigma^{-1}(c)}. \tag{2}$$

The combined group $\mathcal{G} = \mathcal{O} \times S_6$. The *direct product* of $\mathcal{O}$ and S_6 has order $24 \times 720 = 17{,}280$. An element $(g, \sigma) \in \mathcal{G}$ acts on states as

$$[(g, \sigma) \cdot \mathbf{x}]_{i,c} = x_{g^{-1}(i),\, \sigma^{-1}(c)}. \tag{3}$$

Because the cube's distance to the solved state is unchanged by rotating the physical cube or relabeling all colors, $d(\mathbf{x}) = d((g, \sigma) \cdot \mathbf{x})$ for all $(g, \sigma) \in \mathcal{G}$.

Definition 2.3 (Equivariant and invariant maps)**.** *A map $f : X \to Y$ is G-*equivariant *if $f(g \cdot x) = g \cdot f(x)$ for all $g \in G$, $x \in X$, where G acts on both X and Y. If G acts trivially on Y (i.e., $g \cdot y = y$), then $f(g \cdot x) = f(x)$ and f is called G-*invariant*.*

The value network $f : \mathbb{R}^{144} \to \mathbb{R}$ should be $\mathcal{G}$-invariant.

2.3 Equivariant Neural Networks

Equivariant networks build invariance or equivariance directly into the architecture by constraining layer weights via symmetry. The key idea, originating with group-equivariant CNNs [3], is to use weight sharing across orbits: parameters shared across all group-equivalent input-output pairs. This simultaneously reduces the effective parameter count and guarantees exact symmetry.

Pair orbits. For a group G acting on an index set $\{1, \ldots, n\}$, the *pair orbits* are the equivalence classes of pairs (i, j) under

$$(i, j) \sim (i', j') \iff \exists g \in G : g(i) = i',\ g(j) = j'. \tag{4}$$

The number of pair orbits determines the number of independent weight parameters in a G-equivariant linear layer. For $\mathcal{O}$ acting on 24 sticker positions, there are $n_{\text{sp}} = 24$ spatial pair orbits. For S_6 acting on 6 colors, there are only $n_{\text{co}} = 2$ color pair orbits: the orbit of diagonal pairs (c, c) and the orbit of off-diagonal pairs (c, c') with $c \neq c'$.

3 Related Work

Neural solvers for Rubik's cubes. Agostinelli et al. [1] introduced DeepCubeA, which trains a value network via autodidactic iteration (a form of self-supervised reinforcement learning) and uses it to guide A* search. Although DeepCubeA solves the 3×3×3 cube, it achieves only 60.3% solution optimality and requires substantial computation. McAleer et al. [8] improved upon this with approximate policy iteration. These works establish that a learned heuristic can successfully guide search for combinatorial puzzles, but they do not exploit the cube's symmetry structure in the architecture.

Beam search with learned heuristics. The CayleyPy project [2] is the most directly relevant recent work to our evaluation methodology. Chervov et al. propose a general framework in which a neural network estimates a "diffusion distance" to the solved state, and beam search of width W uses these estimates to maintain the W most promising frontier states at each step. They demonstrate that this combination achieves 98.4% optimality on the 3×3×3 cube (versus 60.3% for DeepCubeA) and scales to cubes up to 5×5×5—well beyond the reach of any prior method. A key finding is that solution quality improves approximately linearly with $\log W$, suggesting that heuristic accuracy governs the fundamental efficiency of the approach: a more accurate value function allows a narrower beam to succeed. Chervov et al. explicitly note that *"the neural network provides heuristics on where to make the next steps, while the graph pathfinding technique compensates for any possible incorrectness in the neural network predictions"*, highlighting that beam search is robust to moderate heuristic error but ultimately bottlenecked by heuristic consistency at the local level. This motivates our use of beam search solve rate—rather than MAE or Pearson r alone—as the primary metric for comparing value network quality: a consistent, equivariant value landscape should translate directly into higher solve rates even under a narrow beam.

Equivariant networks for combinatorial structures. Cohen and Welling [3] established the general framework of group-equivariant CNNs. Finzi et al. [4] provide a practical implementation for arbitrary matrix groups via the EMLP framework. Kondor and Trivedi [6] show that equivariance to compact groups can be enforced through convolution on the group. Maron et al. [7] study equivariant networks for sets and symmetric structures more generally. Our work applies these ideas to the specific symmetry group of the Pocket Cube and introduces a novel weight-sharing scheme for the product group $\mathcal{O} \times S_6$.

4 Method

4.1 Spatial Group Convolution Layer (SpatialConv)

The SpatialConv layer maps a hidden state of shape $(B, 24, c_{\text{in}})$ to $(B, 24, c_{\text{out}})$, where B is the batch dimension and the 24-position axis carries the sticker representation.

Weight sharing. Let $\text{orb}_{\text{sp}} : \{1, \ldots, 24\}^2 \to \{1, \ldots, n_{\text{sp}}\}$ map each position pair (i, j) to its orbit index under $\mathcal{O}$. The layer has a weight tensor $\mathbf{W} \in \mathbb{R}^{c_{\text{out}} \times c_{\text{in}} \times n_{\text{sp}}}$ and a bias $\boldsymbol{\beta} \in \mathbb{R}^{c_{\text{out}}}$. The full 24×24 kernel is obtained by index-gathering:

$$K_{o,c,i,j} = W_{o,\, c,\, \text{orb}_{\text{sp}}(i,j)}, \qquad o \in [c_{\text{out}}],\ c \in [c_{\text{in}}],\ i, j \in [24]. \tag{5}$$

Forward pass.

$$y_{b,i,o} = \sum_{j=1}^{24} \sum_{c=1}^{c_{\text{in}}} K_{o,c,i,j} \cdot x_{b,j,c} + \beta_o. \tag{6}$$

Parameter count. $c_{\text{out}} \times c_{\text{in}} \times 24 + c_{\text{out}}$, compared to $c_{\text{out}} \times c_{\text{in}} \times 24^2$ for an unconstrained layer of the same width — a $24\times$ reduction.

4.2 Color-Equivariant Convolution Layer (ColorConv)

The ColorConv layer is the main technical contribution of this paper. It extends SpatialConv to the full group $\mathcal{G} = \mathcal{O} \times S_6$.

Block-of-six channel structure. To enable S_6 to act naturally on the hidden representation, we organize channels in blocks of six: a hidden state with k blocks has $6k$ total channels, grouped as a tensor of shape $(B, 24, k, 6)$, where the last axis carries a 6-dimensional representation of S_6. Under $(g, \sigma) \in \mathcal{G}$, the hidden state transforms as

$$[(g, \sigma) \cdot \mathbf{h}]_{b,i,\kappa,d} = h_{b,\, g^{-1}(i),\, \kappa,\, \sigma^{-1}(d)}. \tag{7}$$

Color pair orbits. Under the conjugation action of S_6 on pairs $(d, c) \in \{1, \ldots, 6\}^2$, there are exactly two orbits:

$$\text{orb}_{\text{co}}(d, c) = \begin{cases} 0 & \text{if } d = c \quad \text{(same color, 6 pairs)}, \\ 1 & \text{if } d \neq c \quad \text{(different colors, 30 pairs)}. \end{cases} \tag{8}$$

This follows from the fact that S_6 acts transitively on all same-color pairs and on all different-color pairs separately.

Weight tensor and kernel expansion. The layer has weight $\mathbf{W} \in \mathbb{R}^{k_{\text{out}} \times k_{\text{in}} \times n_{\text{sp}} \times 2}$ and bias $\boldsymbol{\beta} \in \mathbb{R}^{k_{\text{out}}}$. The full $24 \times 24 \times 6 \times 6$ kernel is:

$$K_{o,\kappa,i,j,d,c} = W_{o,\, \kappa,\, \text{orb}_{\text{sp}}(i,j),\, \text{orb}_{\text{co}}(d,c)}. \tag{9}$$

Forward pass.

$$y_{b,i,o,d} = \sum_{j=1}^{24} \sum_{\kappa=1}^{k_{\text{in}}} \sum_{c=1}^{6} K_{o,\kappa,i,j,d,c} \cdot x_{b,j,\kappa,c} + \beta_o. \tag{10}$$

The output is then reshaped to $(B, 24, 6\,k_{\text{out}})$ for the next layer.

Parameter count. $k_{\text{out}} \times k_{\text{in}} \times n_{\text{sp}} \times 2 + k_{\text{out}}$. With $n_{\text{sp}} = 24$ and channel width $6k$, this is $24 \times 2 = 48$ independent weights per block pair — compared to $(6k)^2 \times 24^2 = 20{,}736k^2$ for an unconstrained layer, a compression of **0.23%** (for any k).

4.3 Comparison with the General EMLP Framework

Both our layers and the general EMLP framework of Finzi et al. [4] solve the same underlying mathematical problem: characterising the space of G-equivariant linear maps $\text{Hom}_G(V, W)$ between two group representations. They differ in how that space is computed.

EMLP (Finzi et al.) — numerical approach. For a group G with generators $\{g_i\}$, the equivariance constraint $L\,\rho_{\text{in}}(g) = \rho_{\text{out}}(g)\,L$ on a linear map $L \in \mathbb{R}^{|W|\times|V|}$ can be written as

$$C\,\text{vec}(L) = \mathbf{0}, \qquad C = \bigoplus_i \Big(\rho_{\text{in}}(g_i)^\top \otimes I_{|W|} - I_{|V|} \otimes \rho_{\text{out}}(g_i)\Big). \tag{11}$$

The equivariant basis $\{B_k\}$ is the **null space of** C, found via Krylov iterations or SVD. The layer is then parameterised as $L = \sum_k c_k B_k$ with learnable scalars $\{c_k\}$. This approach is fully general: it applies to any group (including continuous groups such as SO(3) or SE(3)) and any representation, at the cost of constructing and factorising C, which has size $O(|V|^2|W|^2)$.

Orbit enumeration (our approach) — combinatorial approach. For a finite group acting by permutation, the null space of C has a direct combinatorial description: by Schur's lemma, the entry L_{ij} depends only on the orbit of the pair (i, j) under the group action. We therefore compute orbit indices directly by exhaustive enumeration of the group,

$$\text{orb}(i, j) = \text{orb}(i', j') \iff \exists\, g \in G : g(i) = i',\ g(j) = j', \tag{12}$$

and parameterise $L_{ij} = W_{\text{orb}(i,j)}$ with one learnable scalar per orbit. The basis vectors are the orbit indicator matrices $[B_k]_{ij} = \mathbf{1}[\text{orb}(i, j) = k]$, which are exactly the null-space basis that EMLP would recover numerically.

Equivalence and trade-offs. For finite permutation representations the two approaches produce identical equivariant function classes: the null space of C equals the span of the orbit indicator matrices. For the 2×2×2 cube, $G = \mathcal{O}$ has only 24 elements acting on 24 sticker positions. Orbit enumeration computes all pair orbits in milliseconds with zero floating-point error, making EMLP's Krylov machinery unnecessary. The general EMLP framework becomes the right tool when the group is continuous (e.g. learning a value function equivariant to SO(3) rotations in 3D space) or when the representation is not a simple permutation action.

4.4 Invariant Pooling Heads

Both networks end with an invariant pooling head that reduces the equivariant representation to a scalar.

Spatial pooling (for EMLP). Average over the 24 position axis, then apply a linear projection:

$$f(\mathbf{x}) = \mathbf{w}^\top \left(\frac{1}{24} \sum_{i=1}^{24} \mathbf{h}_i \right) + b, \qquad \mathbf{w} \in \mathbb{R}^{c_{\text{hidden}}}. \tag{13}$$

Spatial + color pooling (for EMLP-Col). Average over both the 24 positions and the 6 colors within each block, then apply a linear projection:

$$f(\mathbf{x}) = \mathbf{w}^\top \left(\frac{1}{144} \sum_{i=1}^{24} \sum_{d=1}^{6} \mathbf{h}_{i,\cdot,d} \right) + b, \qquad \mathbf{w} \in \mathbb{R}^{k_{\text{hidden}}}. \tag{14}$$

4.5 Full Model Architectures

Both equivariant models use 3 hidden layers with ReLU activations. Table 1 summarizes the three architectures compared in this work.

Table 1: The three model architectures compared in this work. All models use 3 hidden layers with ReLU activations. Parameter counts use widths $c = 28$ (EMLP), $k = 20$ (EMLP-Col), $h = 110$ (MLP). †Two additional baselines were evaluated but are not the primary focus: an augmented MLP (MLP-Aug, $h = 110$, ∼40K) and a large unconstrained MLP ($h = 256$, ∼169K); see footnote in Table 2.

Model	Layer type	Width	Params	Symmetry	Role
EMLP	SpatialConv	$c = 28$	∼42K	$\mathcal{O}$	Spatial equivariance
EMLP-Col	ColorConv	$k = 20$	∼39K	$\mathcal{O} \times S_6$	Spatial + color equivariance
MLP	Linear	$h = 110$	∼40K	None	Parameter-matched baseline

Input/output. All models accept a flattened one-hot state vector of dimension 144. Equivariant models immediately reshape to $(24, 6)$ (or $(24, 1, 6)$ for EMLP-Col) before processing. All models output a scalar prediction $\hat{d} \in [0, 1]$, which is scaled back to move units as $14 \cdot \hat{d}$.

4.6 Training Setup

Dataset. We precompute exact BFS distances for all 3,674,160 reachable states. The training set of 200,000 samples is drawn with equal probability from each depth $d \in \{1, \ldots, 14\}$ (stratified). The validation set has 20,000 randomly sampled states; the test set has ≈1,000 states at each depth $d \in \{0, \ldots, 14\}$ (≈15K total).

Objective. We minimize mean squared error on normalized targets:

$$\mathcal{L} = \frac{1}{N} \sum_{n=1}^{N} \left(f(\mathbf{x}^{(n)}) - \frac{d(\mathbf{x}^{(n)})}{14} \right)^2. \tag{15}$$

Optimization. We use Adam [5] with initial learning rate 10^{-3}, decayed via cosine annealing to 10^{-5} over 200 epochs. Batch size is 512. Early stopping is applied with patience 15 on validation MSE; the best checkpoint is saved and used for evaluation. All experiments use 3 independent random seeds (42, 43, 44) to estimate variance.

5 Theoretical Analysis

5.1 Equivariance of SpatialConv

Proposition 5.1 (SpatialConv is $\mathcal{O}$-equivariant)**.** *Let $\pi \in \mathcal{O}$ act on sticker positions. For any input $\mathbf{x} \in \mathbb{R}^{B\times 24\times c_{\text{in}}}$,*

$$\textit{SpatialConv}(\pi \cdot \mathbf{x}) = \pi \cdot \textit{SpatialConv}(\mathbf{x}), \tag{16}$$

where $[\pi \cdot \mathbf{x}]_{b,i,c} = x_{b,\pi^{-1}(i),c}$ on the input and $[\pi \cdot \mathbf{y}]_{b,i,o} = y_{b,\pi^{-1}(i),o}$ on the output.

Proof. Fix b, i, o. By definition of pair orbits, for any $\pi \in \mathcal{O}$,

$$\text{orb}_{\text{sp}}\big(\pi^{-1}(i),\, j'\big) = \text{orb}_{\text{sp}}\big(i,\, \pi(j')\big) \tag{17}$$

for all $j' \in \{1, \ldots, 24\}$. (This holds because $(i, \pi(j'))$ and $(\pi^{-1}(i), j')$ are related by the group element π^{-1}: $\pi^{-1}(i) = \pi^{-1}(i)$ and $\pi^{-1}(\pi(j')) = j'$.)

We compute the i-th output of $\textsc{SpatialConv}(\pi \cdot \mathbf{x})$:

$$\begin{aligned}
[\textsc{SpatialConv}(\pi \cdot \mathbf{x})]_{b,i,o} &= \sum_{j=1}^{24} \sum_{c=1}^{c_{\text{in}}} W_{o,c,\,\text{orb}_{\text{sp}}(i,j)} \cdot [\pi \cdot \mathbf{x}]_{b,j,c} \\
&= \sum_{j=1}^{24} \sum_{c=1}^{c_{\text{in}}} W_{o,c,\,\text{orb}_{\text{sp}}(i,j)} \cdot x_{b,\pi^{-1}(j),c}.
\end{aligned} \tag{18}$$

Substituting $j' = \pi^{-1}(j)$ (so $j = \pi(j')$, and the sum still ranges over all 24 positions):

$$\begin{aligned}
&= \sum_{j'=1}^{24} \sum_{c=1}^{c_{\text{in}}} W_{o,c,\,\text{orb}_{\text{sp}}(i,\,\pi(j'))} \cdot x_{b,j',c} \\
&\overset{(17)}{=} \sum_{j'=1}^{24} \sum_{c=1}^{c_{\text{in}}} W_{o,c,\,\text{orb}_{\text{sp}}(\pi^{-1}(i),\,j')} \cdot x_{b,j',c} \\
&= [\textsc{SpatialConv}(\mathbf{x})]_{b,\pi^{-1}(i),o} = [\pi \cdot \textsc{SpatialConv}(\mathbf{x})]_{b,i,o}\,.
\end{aligned}$$

□

5.2 Equivariance of ColorConv

Theorem 5.2 (ColorConv is $\mathcal{G}$-equivariant)**.** *Let $(g, \sigma) \in \mathcal{G} = \mathcal{O} \times S_6$ act as in (7). For any input $\mathbf{x} \in \mathbb{R}^{B\times 24\times k_{\text{in}}\times 6}$,*

$$\textit{ColorConv}((g, \sigma) \cdot \mathbf{x}) = (g, \sigma) \cdot \textit{ColorConv}(\mathbf{x}). \tag{19}$$

Proof. Fix b, i, o, d. We compute the (i, o, d)-th output of $\textsc{ColorConv}((g, \sigma) \cdot \mathbf{x})$:

$$\begin{aligned}
[\textsc{ColorConv}((g, \sigma) \cdot \mathbf{x})]_{b,i,o,d} &= \sum_{j,\kappa,c} K_{o,\kappa,i,j,d,c} \cdot [(g, \sigma) \cdot \mathbf{x}]_{b,j,\kappa,c} \\
&= \sum_{j,\kappa,c} W_{o,\kappa,\text{orb}_{\text{sp}}(i,j),\text{orb}_{\text{co}}(d,c)} \cdot x_{b,g^{-1}(j),\kappa,\sigma^{-1}(c)}.
\end{aligned} \tag{20}$$

Substituting $j' = g^{-1}(j)$ and $c' = \sigma^{-1}(c)$:

$$= \sum_{j',\kappa,c'} W_{o,\kappa,\,\text{orb}_{\text{sp}}(i,\,g(j')),\,\text{orb}_{\text{co}}(d,\,\sigma(c'))} \cdot x_{b,j',\kappa,c'}. \tag{21}$$

We now simplify the two orbit indices.

Spatial orbit: By the same argument as Proposition 5.1, $\mathrm{orb}_{\mathrm{sp}}(i, g(j')) = \mathrm{orb}_{\mathrm{sp}}(g^{-1}(i), j')$.

Color orbit: Since $\sigma \in S_6$ is a bijection, $d = \sigma(c')$ if and only if $\sigma^{-1}(d) = c'$. Therefore the equality relation $d = \sigma(c')$ is equivalent to $\sigma^{-1}(d) = c'$, so:

$$\mathrm{orb}_{\mathrm{co}}(d,\, \sigma(c')) = \begin{cases} 0 & \text{if } d = \sigma(c') \\ 1 & \text{if } d \neq \sigma(c') \end{cases} = \begin{cases} 0 & \text{if } \sigma^{-1}(d) = c' \\ 1 & \text{if } \sigma^{-1}(d) \neq c' \end{cases} = \mathrm{orb}_{\mathrm{co}}(\sigma^{-1}(d),\, c'). \tag{22}$$

Substituting both simplifications into (21):

$$\begin{aligned} &= \sum_{j',\kappa,c'} W_{o,\kappa,\,\mathrm{orb}_{\mathrm{sp}}(g^{-1}(i),\,j'),\,\mathrm{orb}_{\mathrm{co}}(\sigma^{-1}(d),\,c')} \cdot x_{b,j',\kappa,c'} \\ &= [\textsc{ColorConv}(\mathbf{x})]_{b,\,g^{-1}(i),\,o,\,\sigma^{-1}(d)} \\ &= [(g,\sigma) \cdot \textsc{ColorConv}(\mathbf{x})]_{b,i,o,d}\,. \end{aligned}$$

□

5.3 Invariance of the Full Networks

Corollary 5.3 (Full networks are $\mathcal{G}$-invariant)**.** *Let Φ_{emlp} and Φ_{col} denote the full EMLP and EMLP-Col networks respectively. For all $(g,\sigma) \in \mathcal{G}$ and all states $\mathbf{x}$:*

$$\Phi_{\mathrm{col}}((g,\sigma)\cdot\mathbf{x}) = \Phi_{\mathrm{col}}(\mathbf{x}), \qquad \Phi_{\mathrm{emlp}}(g\cdot\mathbf{x}) = \Phi_{\mathrm{emlp}}(\mathbf{x}). \tag{23}$$

Proof. Each hidden layer is equivariant by Theorem 5.2 (resp. Proposition 5.1). The composition of equivariant maps is equivariant. The pooling heads (Equations (13) and (14)) produce an output that is invariant: they average over the group orbit of each hidden state and apply a weight vector that is itself independent of position and color. Formally, averaging over all positions (and colors) eliminates the dependence on the particular representative chosen, yielding a scalar that satisfies $f(g\cdot\mathbf{x}) = f(\mathbf{x})$. □

5.4 Implicit Data Augmentation

Proposition 5.4 (Implicit $|\mathcal{G}|$-fold augmentation)**.** *Suppose f is $\mathcal{G}$-invariant and trained to minimize $\mathcal{L}(f(\mathbf{x}), y)$ on a dataset $\mathcal{D} = \{(\mathbf{x}^{(n)}, y^{(n)})\}$. Then for every training example $(\mathbf{x}^{(n)}, y^{(n)}) \in \mathcal{D}$, f automatically satisfies*

$$\mathcal{L}\Big(f\Big((g,\sigma)\cdot\mathbf{x}^{(n)}\Big),\, y^{(n)}\Big) = \mathcal{L}\Big(f(\mathbf{x}^{(n)}),\, y^{(n)}\Big) \tag{24}$$

for all $(g,\sigma) \in \mathcal{G}$—a total of 17,280 *equivalent training examples per sample, with no additional computation.*

Proof. Direct from the definition of $\mathcal{G}$-invariance: $f((g,\sigma)\cdot\mathbf{x}^{(n)}) = f(\mathbf{x}^{(n)})$. □

Remark 5.5. *Explicit data augmentation (as in MLP-Aug) randomly samples a subset of $\mathcal{G}$ each iteration and applies the transformations to the batch. This provides approximate coverage of $\mathcal{G}$ in expectation, but is strictly weaker than built-in equivariance in three ways: (1) coverage is stochastic, not exact; (2) a finite-capacity model may fail to generalize from augmented examples; (3) each augmented example consumes forward/backward pass computation, whereas equivariance is free at inference time.*

5.5 Parameter Efficiency

The weight tensor of ColorConv has $k_{\text{out}} \times k_{\text{in}} \times n_{\text{sp}} \times 2$ parameters. An unconstrained linear layer with the same input and output dimensionality $(6k_{\text{in}}) \times (6k_{\text{out}})$ over 24 positions (i.e., a standard MLP operating on the flattened 24-position, 6-color, k-block feature) would require $6k_{\text{out}} \times 6k_{\text{in}} \times 24 \times 24 = 20{,}736\, k_{\text{out}}k_{\text{in}}$ parameters. The compression ratio is:

$$\frac{k_{\text{out}}k_{\text{in}} \times 24 \times 2}{6k_{\text{out}} \times 6k_{\text{in}} \times 24 \times 24} = \frac{48}{20{,}736} \approx 0.23\%. \tag{25}$$

This extreme compression is possible because $\mathcal{G}$ has a large orbit structure relative to the state dimension, leaving very few degrees of freedom in the equivariant function class.

6 Experiments

6.1 Experimental Design

Our experiment is a controlled comparison: we fix the training data (200K stratified samples), optimizer (Adam with cosine schedule), network depth (3 hidden layers with ReLU), and approximate parameter count ($\sim$40K) across all three models. The only variable that differs between models is the symmetry constraint imposed on the weight tensor: none (MLP), spatial rotations (EMLP), or spatial rotations plus color permutations (EMLP-Col). This isolation allows us to attribute performance differences directly to the equivariance structure rather than to model capacity, optimization, or data quantity. Each configuration is trained with 3 independent random seeds (42, 43, 44) to estimate variance.

6.2 Evaluation Metrics

We evaluate each model on the held-out test set ($\approx$15K states, $\sim$1000 per depth) using the following metrics:

- **MAE**: Mean absolute error $\frac{1}{N}\sum |14 \cdot \hat{d}_n - d_n|$ in move units.
- **Rounded accuracy**: Fraction of test states where $\lfloor 14 \cdot \hat{d} + 0.5 \rfloor = d$.
- **Pearson** r: Correlation between predicted and true distances.
- **Equivariance error**: Mean $|f(g \cdot \mathbf{x}) - f(\mathbf{x})|$ over random $(g, \sigma) \in \mathcal{G}$ and test states; measures how closely a model satisfies the symmetry constraint empirically.
- **Greedy solve rate**: Fraction of scrambled cubes solved by beam search (beam width $W = 20$, max 50 moves) guided by f. We adopt beam search as the primary evaluation criterion following Chervov et al. [2], who show that beam search with a learned heuristic is the most effective strategy for neural cube solving, and that heuristic quality—specifically its local consistency across symmetry-equivalent states—is the key bottleneck. A narrow beam ($W = 20$) is intentionally chosen to stress-test value network quality: an inaccurate or inconsistent heuristic will fail immediately, while a well-calibrated equivariant heuristic should succeed even under this constraint.

Table 2: Main results on the 200K training set (seed 42), mean over 3 seeds. Best result per column in **bold**. *[pending]*: MLP results will be filled in once training completes. ‡ Supplementary baselines not part of the primary comparison: MLP-Aug ($h = 110$, augmented, ∼40K params): MAE 2.36, r 0.36, solve rate 1.4% at depth 10; MLP-Large ($h = 256$, unconstrained, ∼169K params): MAE 1.32, r 0.81, solve rate 0.0% at depth 10.

Model	**Params**	**MAE**↓	**Rounded acc.**↑	**Pearson** r↑	**Equiv. err. (moves)**		**Solve rate**	
					spatial	*color*	*depth 7*	*dep*
EMLP-Col	39K	**0.932**	**45.4%**	**0.900**	7.8×10^{-7}	8.3×10^{-7}	**100.0%**	**97**
EMLP	42K	1.239	31.9%	0.839	4.4×10^{-7}	—	**100.0%**	9
MLP	40K	*[pending]*	*[pending]*	*[pending]*	*[pending]*	—	*[pending]*	*[pen*

6.3 Main Results

6.4 Key Findings

Finding 1: Exact equivariance is achieved in practice. Both equivariant models achieve spatial equivariance errors of $\sim10^{-7}$ moves — at the limit of float32 numerical precision — confirming that the orbit-based weight sharing enforces exact invariance at runtime, not merely in expectation. EMLP-Col additionally achieves color equivariance error of 8.3×10^{-7} moves, confirming joint $\mathcal{O}\times S_6$ invariance. The parameter-matched MLP must learn any symmetry from data; its equivariance error (*pending*) is expected to be orders of magnitude larger, reflecting the inconsistency that harms beam search.

Finding 2: Symmetry vs. no symmetry at equal parameters. The central comparison of this paper is EMLP and EMLP-Col (both ∼40K parameters, built-in symmetry) against the parameter-matched MLP (∼40K parameters, no symmetry). Supplementary results reinforce this picture: a large unconstrained MLP with 4× more parameters (∼169K) achieves Pearson $r = 0.81$ yet a 0.0% beam-search solve rate at every depth (see Table 2 footnote), demonstrating that global accuracy metrics alone do not predict solver performance. The equivariant models' consistent value landscapes — guaranteed by Corollary 5.3 — provide the per-move ranking reliability that beam search requires.

Finding 3: Color equivariance adds value over spatial-only. At nearly identical parameter counts (39K vs. 42K), EMLP-Col outperforms EMLP on all metrics: MAE (0.932 vs. 1.239 moves), Pearson r (0.900 vs. 0.839), and beam-search solve rate at depth 10 (97.2% vs. 94.2%). The additional 720-element S_6 symmetry, despite acting on only 6 color channels, provides a measurable benefit at negligible extra parameter cost — consistent with Proposition 5.4, which shows that a larger symmetry group yields stronger implicit data augmentation.

6.5 Plots

7 Discussion

Why does the MLP fail at search? The greedy solver makes a sequence of local decisions: at each step, it evaluates six successor states and picks the one with the lowest predicted distance.

[Learning curves]

(a) Validation MSE vs. epoch (200K training set).

[Per-depth MAE]

(b) MAE broken down by BFS depth.

Figure 2: Training dynamics and per-depth prediction error.

[Solve rates]

(a) Greedy solve rate by scramble depth.

[Equivariance error]

(b) Equivariance error by symmetry type (log scale).

Figure 3: Solve performance and symmetry measurement.

Even a model with reasonable global correlation can fail if its predictions are inconsistent across symmetry-equivalent states: two physically identical positions (related by a rotation) may receive different predicted distances, causing the solver to cycle or take suboptimal paths. By contrast, EMLP and EMLP-Col guarantee $f(\pi \cdot \mathbf{x}) = f(\mathbf{x})$ for all $\pi \in \mathcal{O}$ (resp. $(g, \sigma) \in \mathcal{G}$). This consistency is not captured by Pearson r or MAE, but is critical for reliable search. This phenomenon is starkly illustrated by the supplementary baselines: the large MLP ($\sim$169K parameters) achieves Pearson $r = 0.81$ yet a 0% beam-search solve rate at every depth—demonstrating that even 4$\times$ more parameters cannot compensate for the lack of built-in symmetry. Our finding aligns with the characterization in Chervov et al. [2]: beam search "compensates for any possible incorrectness in the neural network predictions," but only when the heuristic maintains consistent local rankings across equivalent states. The implicit 17,280-fold consistency enforced by equivariance provides exactly this guarantee.

The role of color symmetry. The S_6 group is substantially larger than $\mathcal{O}$ (720 vs. 24 elements), yet acts on a low-dimensional space (6 colors), resulting in only 2 color pair orbits. This makes the ColorConv layer remarkably compact: the color symmetry is captured by a single binary parameter (same vs. different color) per spatial orbit. The gain over spatial-only EMLP suggests that color relabeling invariance is a non-trivial inductive bias for this problem, even though all six colors are symmetric in the training data.

Threats to validity. Several factors limit the generalizability of our conclusions:

- **Single puzzle.** We evaluate only the 2$\times$2$\times$2 cube. While this provides exact ground-truth distances, the small state space (3.7M states) may overstate the benefits of equivariance

compared to larger cubes where data coverage is necessarily sparser.

- **Limited seeds.** Three random seeds (42, 43, 44) provide a rough variance estimate but not statistically rigorous confidence intervals.
- **Fixed hyperparameters.** We use the same optimizer (Adam), learning rate schedule, and depth (3 layers) for all models. The MLP may benefit from a different hyperparameter configuration (e.g., deeper networks, dropout, or different learning rates); we did not perform a hyperparameter search per model type.
- **Beam width.** A narrow beam ($W = 20$) intentionally stress-tests heuristic quality, but it means that a sufficiently wide beam might close the gap between equivariant and unconstrained models.
- **Supplementary baselines.** The augmented MLP-Aug ($h = 110$) may be affected by the interaction between small capacity and heavy augmentation (sampling from $|\mathcal{G}| = 17{,}280$ transformations), making it difficult to isolate the effect of augmentation alone from that result.

Future work. The equivariant architecture scales naturally to larger cubes: the 3×3×3 cube has a larger symmetry group and state space, where data efficiency gains should be even more pronounced. Incorporating the equivariant value network into tree search (A*, MCTS) would allow evaluation on harder scrambles and comparison with state-of-the-art solvers such as CayleyPy [2]. A hyperparameter search per model type would strengthen the fairness of the comparison, as would evaluating across multiple training set sizes to characterize data efficiency more precisely.

8 Conclusion

We presented two group-equivariant value network architectures for the 2×2×2 Pocket Cube. EMLP encodes invariance to the 24-element spatial rotation group $\mathcal{O}$; EMLP-Col extends this to the full 17,280-element symmetry group $\mathcal{G} = \mathcal{O} \times S_6$ via a novel block-of-six channel structure. We formally proved that both architectures are exactly invariant to their respective symmetry groups, and established that equivariance is equivalent to implicit 17,280-fold data augmentation with zero computational overhead.

In a controlled comparison at equal parameter count (∼40K), both equivariant models substantially outperform the unconstrained MLP baseline, with EMLP-Col achieving a 97.2% greedy solve rate at scramble depth 10 and EMLP achieving 94.2%. A key insight is that equivariance matters for local consistency of the value landscape, not just average prediction accuracy: a supplementary large MLP with 4× more parameters achieves Pearson $r = 0.81$ yet 0% solve rate, while equivariant models succeed reliably.

These results suggest that explicitly encoding known symmetry structure into neural network architectures is particularly beneficial for search-based combinatorial reasoning, where local decision quality matters at every step.

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